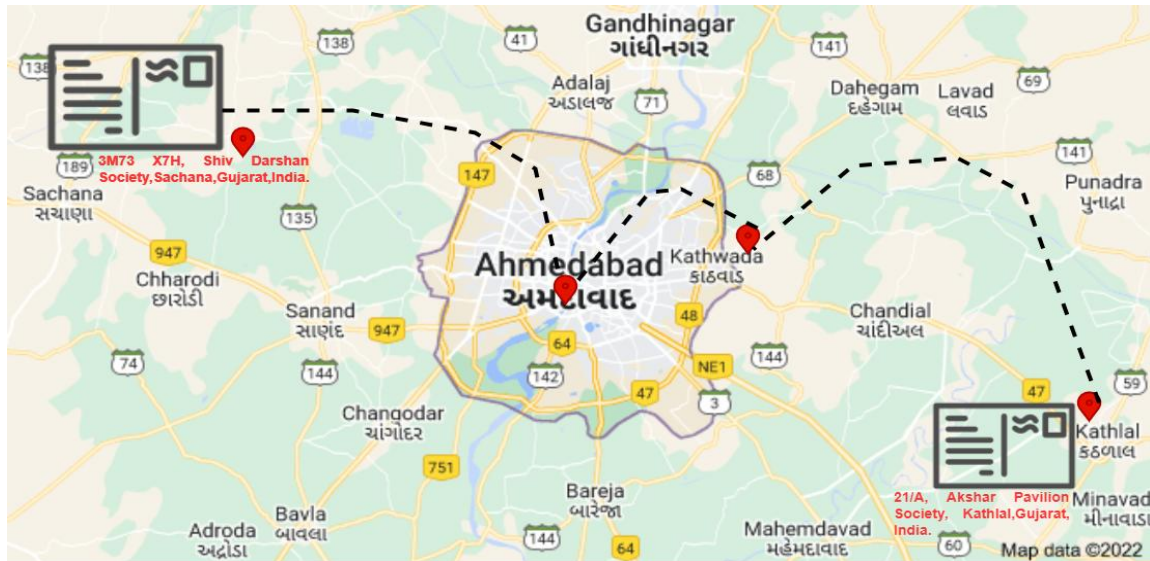


Date:

EXPERIMENT NO. 3**AIM: Allocating IP address to network topologies.****Refer to the following scenarios and let's understand What is an IP Address and Purpose of IP Address.****Scenario : 1**

In this first scenario, an envelope is sent from sachana to kathlall. But there was some check point like ahmedabad, kathwada and then the envelope is delivered to kathlall.

Justify the following statement.

1. Was there any difficulty faced during sending the envelope from sender to receiver?

Scenario : 2

In this second scenario, the sender was located at Ahmedabad and wanted to send a file to the receiver who was located at Gandhinagar. But at the time of sending a file to the receiver, the receiver's machine was powered off. So, can the receiver receive the file from the sender or not?

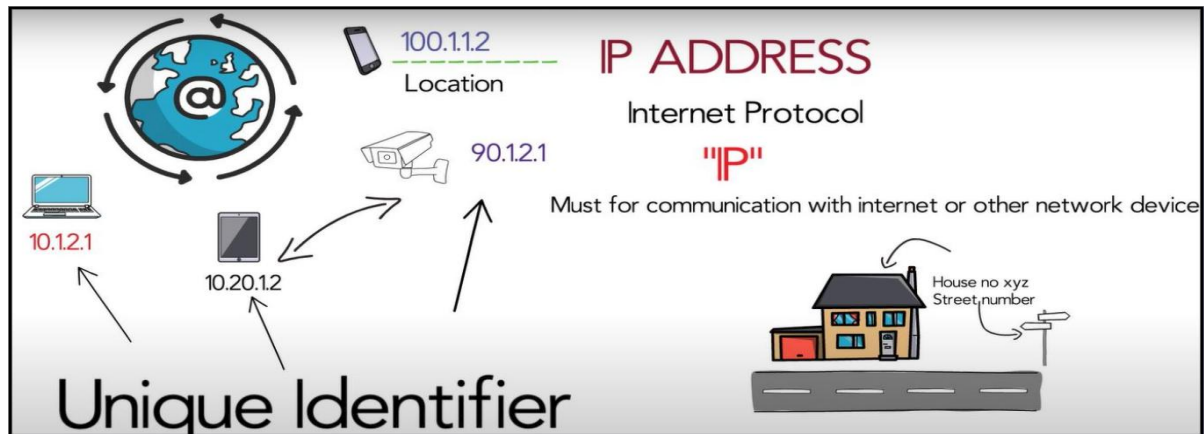
Justify the following statement.

2. Was there any difficulty faced during sending the file from sender to receiver? If yes then what would be the solution for it.

Scenario: 3

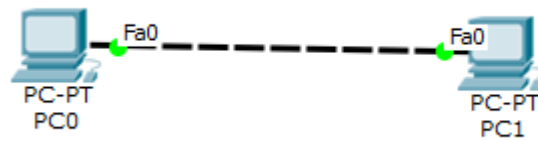
Refer the above image, and justify your answer

Question: Can the envelope be sent to the proper destination?

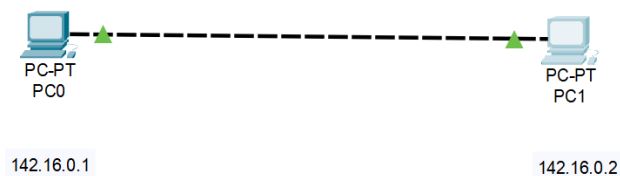
Scenario: 4

An IP address also acts like a return address on postal mail. When a letter you've mailed is delivered to the wrong address, you get the letter back if you include a return address on the envelope. The same holds true for email. When you write to an invalid recipient (such as someone who left their job and no longer has a company email address) your IP address lets the company's mail server send you back a bounce message so that you know your email wasn't sent to the right place.

From the above all scenarios, what would be the conclusion?

Exercise-1(Note: Start allocation IP address number from PC0)

Redraw above diagram which includes IP address and MAC address. Take IP address and MAC address as per your knowledge.



- Ipconfig: fill table ipconfig of all computers
PC0

Parameter	Value
Link local IPV6 Address	FE80::201:64FF:FE18:869E
IP address	142.16.0.1
Subnet Mask	255.255.0.0
Default Gateway	0.0.0.0

PC1

Parameter	Value
Link local IPV6 Address	FE80::2D0:D3FF:FEC7:654C
IP address	142.16.0.2
Subnet Mask	255.255.0.0
Default Gateway	0.0.0.0

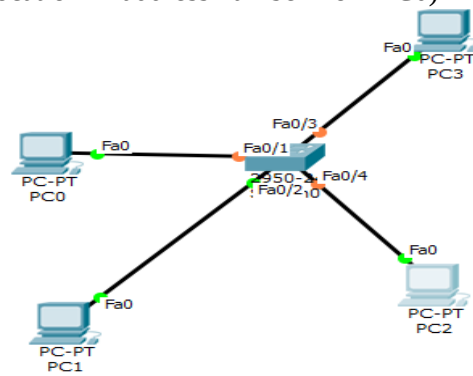
- Ipconfig /all: apply command on command prompt and write parameters and values in the following table.

PC0

Parameter	Value
Physical Address	0001.6418.869E
Link-Local IPv6 Address	FE80::201:64FF:FE18:869F
IPv4 Address	142.16.0.1
Subnet Mask	255.255.0.0
Default Gateway	0.0.0.0
DHCP Servers	0.0.0.0
DHCPv6 Client DUID	00-01-00-01-CE-DD-3D-77-00-01-64-18-86-9E
DNS Servers	0.0.0.0

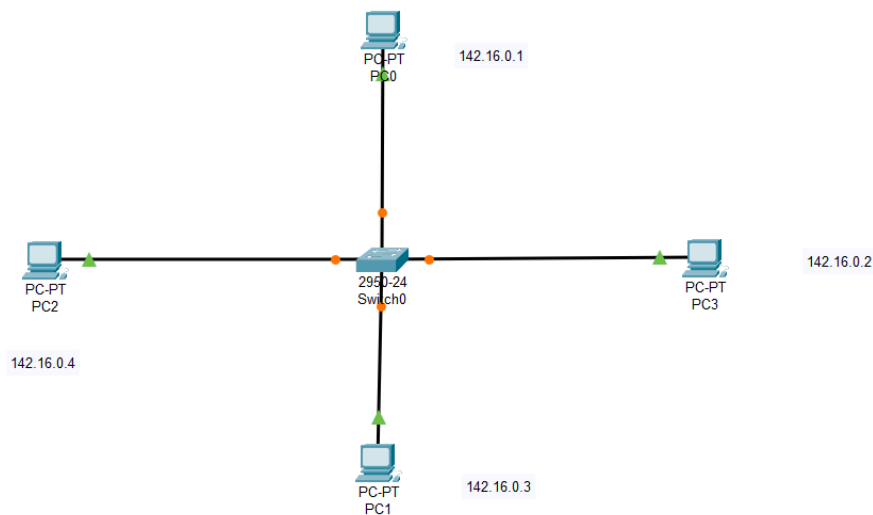
PC1

Parameter	Value
Physical Address	00D0.D3C7.654C
Link-Local IPv6 Address	FE80::2D0:D3FF:FEC7:654C
IPv4 Address	142.16.0.2
Subnet Mask	255.255.0.0
Default Gateway	0.0.0.0
DHCP Servers	0.0.0.0
DHCPv6 Client DUID	00-01-00-01-91-1C-97-B6-00-D0-D3-C7-65-4C
DNS Servers	0.0.0.0

Exercise-2: (Note: Start allocation IP address number from PC0)

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Redraw above diagram which includes IP address and MAC address. Take IP address and MAC address as per your instruction.



Ipconfig: fill table ipconfig of all computers

PC0

Link local IPV6 Address	FE80::20C:CFFF:FE7A:4B6D
IP address	142.16.0.1
Subnet Mask	255.255.0.0
Default Gateway	0.0.0.0

PC1

Link local IPV6 Address	FE80::2D0:FFFF:FEA8:E0E8
IP address	142.16.0.3

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Subnet Mask	255.255.0.0
Default Gateway	0.0.0.0

PC2

Link local IPV6 Address	FE80::203:E4FF:FE4B:29C9
IP address	142.16.0.4
Subnet Mask	255.255.0.0
Default Gateway	0.0.0.0

PC3

Link local IPV6 Address	FE80::201:64FF:FE01:8B6E
IP address	142.16.0.2
Subnet Mask	255.255.0.0
Default Gateway	0.0.0.0

- Ipconfig /all: apply command on command prompt and write parameters and values in following table.

PC0

Parameter	Value
Physical Address	000C.CF7A.4B6D
Link-Local IPv6 Address	FE80::20C:CFFF:FE7A:FE7A:4B6D
IPv4 Address	142.16.0.1
Subnet Mask	255.255.0.0
Default Gateway	0.0.0.0
DHCP Servers	0.0.0.0
DHCPv6 Client DUID	00-01-00-01-64-E9-C4-71-00-0C-CF-7A-4B-6D
DNS Servers	0.0.0.0

PC1

Parameter	Value
Physical Address	00D0.FFA8.E0E8
Link-Local IPv6 Address	FE80::2D0:FFFF:FEAS:E0E8
IPv4 Address	142.16.0.3

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CE262-Data Communication and Networking

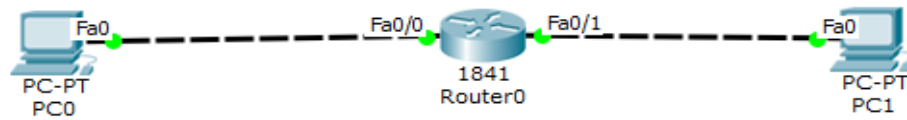
Subnet Mask	255.255.0.0
Default Gateway	0.0.0.0
DHCP Servers	0.0.0.0
DHCPv6 Client DUID	00-01-00-01-56-82-B5-DA-00-D0-FF-A8-E0-E8
DNS Servers	0.0.0.0

PC2

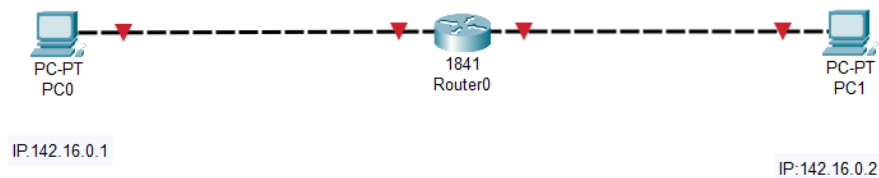
Parameter	Value
Physical Address	0003.E44B.29C9
Link-Local IPv6 Address	FE80::203:E4FF:FE4B:29C9
IPv4 Address	142.16.0.4
Subnet Mask	255.255.0.0
Default Gateway	0.0.0.0
DHCP Servers	0.0.0.0
DHCPv6 Client DUID	00-01-00-01-64-E9-C4-71-00-0C-CF-7A-4B-6D
DNS Servers	0.0.0.0

PC3

Parameter	Value
Physical Address	000C.CF7A.4B6D
Link-Local IPv6 Address	FE80::201:64FF:FE01:8B6E
IPv4 Address	142.16.0.2
Subnet Mask	255.255.0.0
Default Gateway	0.0.0.0
DHCP Servers	0.0.0.0
DHCPv6 Client DUID	00-01-00-01-83-88-D8-DE-00-01-64-01-8B-6E
DNS Servers	0.0.0.0

Exercise-3 (Note: Start allocation IP address number from PC0)

Redraw above diagram which includes IP address and MAC address. Take IP address and MAC address as per your instruction.



- Ipconfig: fill following table with output of ipconfig of computer.

PC0

Link local IPV6 Address	FE80::201:64FF:FE44:C8E0
IP address	142.16.0.1
Subnet Mask	255.255.0.0
Default Gateway	155.13.0.1

PC1

Link local IPV6 Address	FE80::260:5CFF:FECD:2BC2
IP address	142.16.0.2
Subnet Mask	255.255.0.0
Default Gateway	155.12.0.2

- Ipconfig /all: apply command on command prompt and write parameters and values in following table

PC0

Parameter	Value
Physical Address	0001.6444.C8E0

Link Local IPv6 Address	FE80::201:64FF:FE44:C8E0
IPv6 Address	
IPv4 Address	142.16.0.1
Subnet Mask	255.255.0.0
Default Gateway	155.13.0.1
DHCP Servers	0.0.0.0
DHCPv6 Client DUID	00-01-00-01-54-8D-E9-4E-00-01-64-44-C8-E0
DNS Servers	0.0.0.0

PC1

Parameter	Value
Physical Address	0060.5CCD.2BC2
Link Local IPv6 Address	FE80::260:5CFF:FECD:2BC2
IPv6 Address	
IPv4 Address	142.16.0.2
Subnet Mask	255.255.0.0
Default Gateway	155.12.0.2
DHCP Servers	0.0.0.0
DHCPv6 Client DUID	00-01-00-01-15-3C-89-7D-00-60-5C-5C-2B-C2
DNS Servers	0.0.0.0

CONCLUSIONS:

Post-Lab Assignment Questions
(Write answers in file pages, do not take print of this page)

1. What are the conclusions of the ipconfig command?

2. What are the conclusions of **ipconfig /all** commands?

3. What are the conclusions of **arp -a** command before ping?

4. What are the conclusions of **netstat -r** command after ping?

5. What is my MAC address?

6. Which network is configured? Static and Dynamic

7. What is my gateway?

8. What is a hostname?
9. What is my IPv6 address?
10. What is ARPA?
11. What is a loopback address?
12. What does Port 80 mean?
13. What is the difference between logical address and physical address?
14. What is NetBIOS?